Zachary S. Siegel

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EDUCATION

Massachusetts Institute of Technology, Cambridge, Massachusetts, USA — Sep 2025 – Present

PhD in Electrical Engineering and Computer Science
Advisors: Leslie P. Kaelbling, Tomás Lozano-Pérez, and Joshua B. Tenenbaum

Princeton University, Princeton, New Jersey, USA — Sep 2021 – May 2025

B.S.E. in Computer Science, Minor in Philosophy
Senior thesis: Modeling Open-Ended Goal Inference; advised by Tom Griffiths and Jacob Andreas

AWARDS & FELLOWSHIPS

- Hertz Foundation Fellowship (William J. and Sharon K. Dally Fellowship) 2026
- National Science Foundation Graduate Research Fellowship 2026
- Princeton University: Summa Cum Laude, Tau Beta Pi, Sigma Xi, and Outstanding Computer Science Independent Work Prize 2025

PUBLICATIONS

[1] **Life After Benchmark Saturation: A Case Study of CORE-Bench**
Nitya Nadgir, Sayash Kapoor, Kangheng Liu, Peter Kirgis, Matilda Orona, Stephan Rabanser, Tilman Bayer, Abhishek Shetty, Yue Ling, Derrick Chan-Sew, Rumi Nakagawa, Saiteja Utpala, **Zachary S. Siegel**, Arvind Narayanan
Second Workshop on Agents in the Wild: Safety, Security, and Beyond at ICML (2026)

[2] **Holistic Agent Leaderboard: The Missing Infrastructure for AI Agent Evaluation**
Sayash Kapoor*, Benedikt Stroebl*, Peter Kirgis, Nitya Nadgir, **Zachary S. Siegel**, Boyi Wei, Tianci Xue, Ziru Chen, Felix Chen, Saiteja Utpala, Franck Ndzomga, Dheeraj Oruganty, Sophie Luskin, Kangheng Liu, Botao Yu, Amit Arora, Dongyoon Hahm, Harsh Trivedi, Huan Sun, Juyong Lee, Tengjun Jin, Yifan Mai, Yifei Zhou, Yuxuan Zhu, Rishi Bommasani, Daniel Kang, Dawn Song, Peter Henderson, Yu Su, Percy Liang, Arvind Narayanan
The Fourteenth International Conference on Learning Representations (ICLR 2026)

[3] **Characterizing the Implicit Bias of Regularized SGD in Rank Minimization**
Tomer Galanti, **Zachary S. Siegel**, Aparna Gupte, Tomaso Poggio
Conference on Parsimony and Learning (CPAL 2025)

[4] **AI Agents That Matter**
Sayash Kapoor*, Benedikt Stroebl*, **Zachary S. Siegel**, Nitya Nadgir, Arvind Narayanan
Transactions on Machine Learning Research (2025)

[5] **CORE-Bench: Fostering the Credibility of Published Research Through a Computational Reproducibility Agent Benchmark**
Zachary S. Siegel, Sayash Kapoor, Nitya Nadgir, Benedikt Stroebl, Arvind Narayanan
Transactions on Machine Learning Research (2025)

[6] **BRIGHT: A Realistic and Challenging Benchmark for Reasoning-Intensive Retrieval**
Hongjin Su*, Howard Yen*, Mengzhou Xia*, Weijia Shi, Niklas Muennighoff, Han-yu Wang, Haisu Liu, Quan Shi, **Zachary S. Siegel**, Michael Tang, Ruoxi Sun, Jinsung Yoon, Sercan O Arik, Danqi Chen, Tao Yu
The Thirteenth International Conference on Learning Representations (ICLR 2025)

[7] **Language Guided Operator Learning for Goal Inference**
Zachary S. Siegel, Jiayuan Mao, Nishanth Kumar, Tianmin Shu, Jacob Andreas
Workshop on Learning Effective Abstractions for Planning @ CORL (2024)

[8] **Learning Grounded Action Abstractions from Language**
Lionel Wong*, Jiayuan Mao*, Pratyusha Sharma*, **Zachary S. Siegel**, Jiahai Feng, Noa Korneev, Joshua B Tenenbaum, Jacob Andreas
The Twelfth International Conference on Learning Representations (ICLR 2024)

[9] **Superimposing height-controllable and animated flood surfaces into street-level photographs for risk communication**
Zachary S. Siegel, Scott A Kulp
Weather and Climate Extremes, Volume 32 (2021)

SELECTED RESEARCH EXPERIENCE

Senior Thesis May 2024 – May 2025
Department of Computer Science, Princeton University
Modeling Open-Ended Goal Inference. Investigated how people infer goals in open-ended environments using Bayesian inference over learned libraries of possible plans. Designed the experimental domain and computational model with Tom Griffiths and Jacob Andreas.

Undergraduate Researcher Nov 2023 – May 2025
Center for Information Technology Policy, Princeton University
Building and Evaluating Agent Benchmarks. Developed CORE-Bench, a benchmark and evaluation system for measuring AI agents on computational reproducibility tasks. Studied methodological failure modes in agent evaluation, contributing to AI Agents That Matter and the Holistic Agent Leaderboard.

Visiting Undergraduate Researcher May 2023 – Sep 2023
Language & Intelligence Group, MIT
Learning Grounded Abstractions from Language. Developed methods for learning symbolic action abstractions from language for task-and-motion planning. Implemented low-level policy learning in ALFRED and integrated language-derived operators into the planning pipeline.

Visiting Undergraduate Researcher May 2022 – Aug 2022
Center for Brains, Minds, and Machines, MIT
Investigating the Low-Rank Bias of Neural Networks. Helped derive a theoretical bound characterizing the implicit low-rank bias induced by stochastic gradient descent and weight decay. Evaluated the effect across neural-network architectures through experiments on learned weight matrices and their singular values.

TEACHING EXPERIENCE

- COS 226 (Algorithms) Precept Assistant — Spring 2022
- MAT 203 (Multivariable Calculus) Course Assistant — Fall 2022
- MAT 204 (Linear Algebra) Course Assistant — Spring 2023
- COS 484 (Natural Language Processing) Course Assistant and Grader — Spring 2024
- COS 426 (Computer Graphics) Course Assistant — Fall 2024
- COS 484 (Natural Language Processing) Course Assistant and Grader — Spring 2025

ACADEMIC SERVICE

- *Reviewer*, Trust-AI-Eval (TAE): Can We Trust AI Evaluation? Workshop — NeurIPS 2026
- Co-organizer, Workshop on Useful and Reliable AI Agents — August 2024

[*CV compiled on 2026-09-15*]